

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457085.6804 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.018 +/- 0.048
Transit depth (Rp/Rs)^2	0.034 +/- 0.18 [%]
Orbital Inclination (inc)	78.8 +/- 2.91
Airmass coefficient 1 (a1)	10.36 +/- 3.63
Airmass coefficient 2 (a2)	-1.31 +/- 0.3
Scatter in the residuals of the lightcurve fit is	38.08 %
Best Comparison Star	#2 - [578, 401]
Optimal Aperture	2.17
Optimal Annulus	3.5
Transit Duration (day)	0.017 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.018 ± 0.048 vs 0.1368 (deviation 1.38σ)
Duration fit vs published	0.017 d vs 0.0758 d (deviation 1.31σ)
Mid-time O–C	0.0 min
Depth SNR	0.2
Residual scatter	38.08 %

Light curve

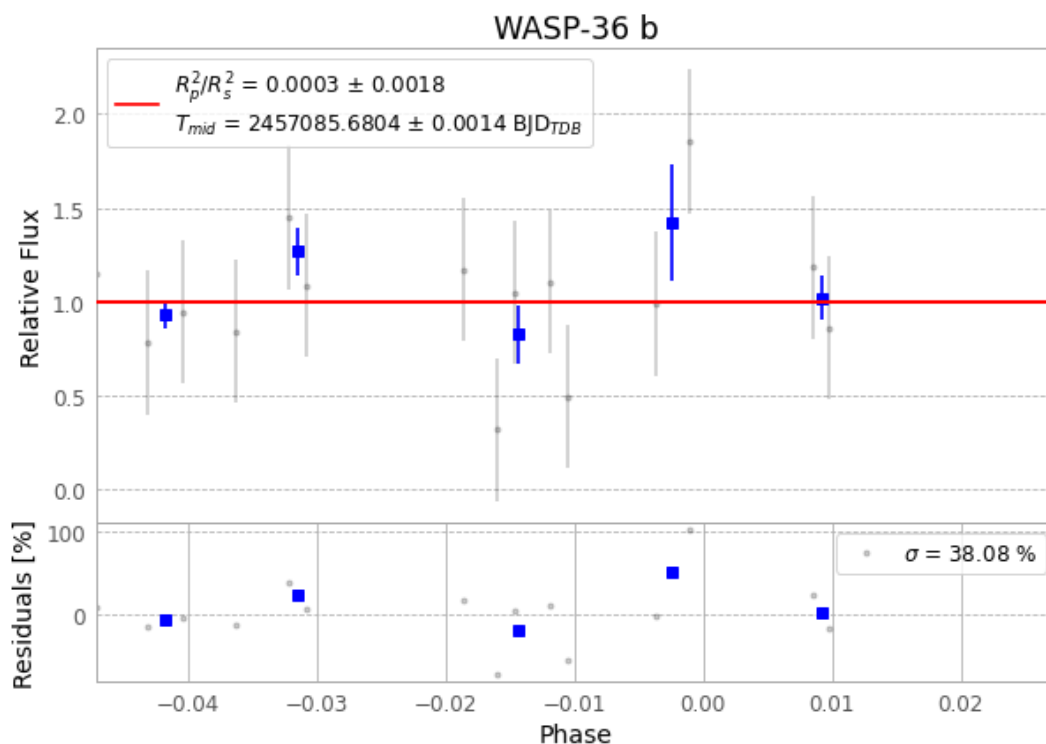


Figure 1 — FinalLightCurve_WASP-36 b_2015-03-03.png (SHA-256 0250daebec11d4b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 376], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 670be705301f4772...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

76 FITS frames (50 MB), WASP-36150304022712.FITS → WASP-36150304061212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-150304.log (6bff0e353bcf...), FinalLightCurve_WASP-36 b_2015-03-03.png (0250daebec11...), FinalLightCurve_WASP-36 b_2015-03-03.csv (a274f2e234d0...)

Compute resources

Wall time 148.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:36Z.